

yt-agent2 Code Review Report

Full-project review spanning agents, UI, database, and Render deployment configuration

1. Executive Summary

This review covers `agents/transcript_agent.py`, `agents/blog_post_agent.py`, `orchestrator.py`, `app.py`, `database.py`, `requirements.txt`, `render.yaml`, `.render-build.sh`, `nixpacks.toml`, `railway.toml`, and `.python-version`. Issues were assessed for syntax, logic, imports, error handling, Render Linux compatibility, 512 MB memory risk, database behaviour, environment variables, and feature-specific correctness.

Severity	Count	Meaning
HIGH	5	Will crash the pipeline, OOM on Render free tier, or leave ffmpeg uninstalled
MEDIUM	8	Incorrect UX, silent data truncation, or degraded reliability
LOW	7	Cleanup, docs drift, dual-platform config confusion
TOTAL	20	Across 11 reviewed files

Critical finding: `_split_audio()` still calls `AudioSegment` after `pydub` was removed from imports and `requirements.txt`. Any audio-download fallback path will raise `NameError` immediately. Separately, `render.yaml` does not invoke `.render-build.sh`, so `ffmpeg` may never be installed on Render even though the shell script exists.

2. Prioritised Issue List

Priority	File	Issue	Impact
HIGH	<code>transcript_agent.py</code>	<code>NameError: AudioSegment</code> used in <code>_split_audio()</code> (L403) but <code>pydub</code> import removed	Audio-download path crashes for every video without captions
HIGH	<code>render.yaml</code> / <code>.render-build.sh</code>	<code>buildCommand</code> is <code>pip install</code> only; <code>.render-build.sh</code> (<code>ffmpeg</code>) is never called	yt-dlp audio conversion fails on Render with <code>ffmpeg</code> missing
HIGH	<code>transcript_agent.py</code>	<code>pydub AudioSegment</code> loads full decoded PCM into RAM for chunking	60-min audio can exceed 512 MB Render free tier and OOM-kill the process
HIGH	<code>transcript_agent.py</code>	<code>_fix_grammar max_tokens=4000</code> truncates long transcripts silently	60-min Whisper output can be cut mid-sentence with no warning
HIGH	<code>transcript_agent.py</code>	Post-download recovery rejects files $\geq$ 24 MB (L233) before chunking	Large successful downloads that raise post-process warnings are retried/lost
MEDIUM	<code>app.py</code>	<code>GROQ_API_KEY</code> warnings still say Railway (L84, L93)	Users on Render follow wrong platform instructions
MEDIUM	<code>app.py</code>	Cache hit ignores uploaded PDF (L123-131)	User expects PDF enrichment but gets stale cached blog post
MEDIUM	<code>transcript_agent.py</code>	<code>_get_transcript_via_api</code> regex misses <code>/shorts/</code> and <code>/embed/</code> (L194)	Captions path skipped for valid YouTube URL formats
MEDIUM	<code>transcript_agent.py</code>	<code>nodejs</code> not installed by Render build; <code>js_runtimes</code> often empty	yt-dlp JS challenge solving unavailable on Render
MEDIUM	<code>app.py</code>	<code>blog_post_to_docx</code> returns <code>b''</code> on failure; download button still shown	User downloads an empty/corrupt Word file with no warning
MEDIUM	<code>database.py</code>	No <code>sslmode</code> enforcement on <code>DATABASE_URL</code> connect	May fail or connect insecurely depending on provider URL shape
MEDIUM	<code>app.py</code>	<code>TranscriptAgent()</code> constructed only for metadata (L103)	Requires <code>GROQ_API_KEY</code> and creates unused Groq client for a metadata call
MEDIUM	<code>transcript_agent.py</code>	Duration guard allows up to 60 min but chunking still broken (<code>AudioSegment</code>)	Videos 20–60 min theoretically supported but crash in practice

Priority	File	Issue	Impact
LOW	railway.toml + render.yaml	Both Railway and Render configs coexist	Confusing which platform is authoritative
LOW	app.py	import re inside if-block (L110)	Style/clarity only; works at runtime
LOW	database.py / app.py	init_db failures swallowed silently (app.py L14-15)	Harder to diagnose missing DATABASE_URL on deploy
LOW	transcript_agent.py	Hardcoded model names whisper-large-v3-turbo / llama-3.3-70b-versatile	Cannot switch models without code change
LOW	transcript_agent.py	Captions path skips grammar fix (intentional but asymmetric)	Caption transcripts never get the stitch/cleanup pass
LOW	nixpacks.toml	Railway-era nixpacks still present after Render migration	Dead config noise in repo
LOW	orchestrator.py	Docstring usage example omits pdf_bytes / pre_fetched_meta	Docs drift from actual API

3. File by File Analysis

3.1 agents/transcript_agent.py

Purpose: Hybrid transcript acquisition — captions API first, then yt-dlp + Whisper with chunking and grammar fix.

Issues:

- **L403 — HIGH:** `AudioSegment.from_mp3(audio_path)` references undefined name after pydub removal. Suggested fix: rewrite `_split_audio()` with `ffmpeg` subprocess (`ffmpeg -i ... -f segment -segment_time 1200`).
- **L397–426 — HIGH:** Even with pydub restored, decoded PCM for a long file can OOM a 512 MB instance. Suggested fix: stream-split via `ffmpeg` without loading the whole file into Python memory.
- **L485 — HIGH:** `max_tokens=4000` on grammar fix. Suggested fix: skip grammar fix for transcripts over ~12k chars, or chunk the fix pass, or raise `max_tokens` and truncate input explicitly.
- **L233 — HIGH:** Recovery path requires size < 24 MB, defeating chunking for large recovered files. Suggested fix: accept any non-empty file and let `_split_audio` / per-chunk checks handle limits.
- **L194 — MEDIUM:** Captions regex lacks `/shorts/` and `/embed/` (present in `app.py`). Align regex with `app.py`.
- **L125–130:** Duration cap is 3600s (60 min). Chunking targets 20-min slices, so videos over 60 minutes are rejected before chunking — correct by current product rule, but the `AudioSegment` crash means 20–60 min videos also fail today.
- **Grammar fix fallback (L491–493):** PASS — returns original transcript on LLM failure.
- **Cookie cleanup (L311–318):** PASS — temp cookie file unlinked in finally.
- **Whisper timeout (L372):** PASS — `timeout=120` present.
- No syntax errors. Imports used: `os`, `re`, `time`, `shutil`, `tempfile`, `yt_dlp`, `Groq`, `dotenv`. **pydub is not imported** (correct removal) but still referenced (incorrect).

3.2 agents/blog_post_agent.py

Purpose: Research + write blog posts via Groq Llama; optional PDF enrichment with relevance gate.

- **PDF relevance (L196–226):** PASS — on failure returns `True` so the pipeline is not blocked.
- **PDF extract (L176–194):** PASS — returns empty string on failure; truncates to 1500 words.
- **Research failure (L66–71):** PASS — falls back to short write prompt.
- No syntax errors. Imports all used (`io`, `os`, `time`, `pdfplumber`, `Groq`, `AuthenticationError`, `dotenv`).
- **Memory note:** Large PDF bytes held in memory once — acceptable with 1500-word text cap.

3.3 orchestrator.py

Purpose: Thin coordinator sequencing `TranscriptAgent` then `BlogPostAgent`.

- No syntax/logic errors. Correctly forwards `pre_fetched_meta` and `pdf_bytes`.
- **LOW:** Class docstring usage example is outdated (omits new parameters).

3.4 app.py

Purpose: Streamlit UI — URL/PDF input, DB cache, pipeline status, regenerate tab.

- **DB cache when metadata fails (L109–114):** PASS — regex extracts `video_id` from `watch/youtu.be/shorts/embed` URLs so cache lookup still runs.
- **Word generation (L18–37):** PASS for crash safety — `try/except` returns `b ' '`. MEDIUM UX gap: empty download still offered.
- **L84, L93 — MEDIUM:** Error copy still references `Railway`; update to `Render`.
- **L123–131 — MEDIUM:** Cache hit short-circuits before PDF is applied. Suggested fix: if PDF uploaded, force regenerate path or bypass cache.
- **Tabs:** Always 3 tabs — avoids `Streamlit ElementNode` crash. Correct.
- No unused imports. No syntax errors.

3.5 database.py

Purpose: PostgreSQL CRUD for cached videos via DATABASE_URL.

- Connection open/close per call is intentional (documented). Fine for single-user.
 - **MEDIUM:** Consider appending `?sslmode=require` when missing from URL for managed Postgres providers.
- Parameterized queries throughout — no SQL injection risk. `ON CONFLICT DO NOTHING` correct.

3.6 requirements.txt

Contents: yt-dlp, groq, python-dotenv, streamlit, python-docx, youtube-transcript-api, psycpg2-binary, pdfplumber.
pydub removed — correct for dependency list, but code still references it.

- All listed packages are compatible with Python 3.12. No Windows-only packages.
- `psycpg2-binary==2.9.9` present — PASS for Render.

3.7 render.yaml

Purpose: Render Blueprint for web service.

- `PYTHONPATH=/opt/render/project/src` — PASS.
- `PYTHON_VERSION=3.12.0` — PASS.
 - **HIGH:** `buildCommand: "pip install -r requirements.txt"` does not run `.render-build.sh`.
Suggested fix: `buildCommand: "./render-build.sh" or bash .render-build.sh` (and ensure executable bit).

3.8 .render-build.sh

Purpose: Install ffmpeg then pip dependencies. Content is correct (`apt-get install -y ffmpeg`). **Issue:** unused until wired into `render.yaml buildCommand`. Must be executable (`chmod +x`).

3.9 nixpacks.toml / railway.toml / .python-version

- `.python-version = 3.12.0` — PASS.
 - `nixpacks.toml / railway.toml` are Railway leftovers. Harmless on Render but add confusion (LOW).
- Railway `startCommand` still uses `PYTHONPATH=/app` — correct for Railway, irrelevant if fully migrated to Render.

4. Render Deployment Checklist

Requirement	Status	Notes
PYTHONPATH=/opt/render/project/src in render.yaml	PASS	Set under envVars
psycopg2-binary in requirements.txt	PASS	psycopg2-binary==2.9.9
ffmpeg installed via .render-build.sh	FAIL	Script exists but buildCommand does not call it
Python 3.12 pinned in .python-version	PASS	3.12.0
Packages compatible with Python 3.12	PASS	No known 3.12 blockers in current pins
pydub still imported anywhere?	PASS*	Import removed; *AudioSegment still referenced — runtime break
GROQ_API_KEY documented for Render	FAIL	app.py UI still says Railway Variables
DATABASE_URL optional with graceful degrade	PASS	init_db / lookups wrapped; app continues
512 MB free-tier memory safety	FAIL	pydub in-memory decode + large audio risk OOM
Streamlit headless + \$PORT binding	PASS	Correct in startCommand

Feature-specific verification

- 1. **Chunking for videos over 60 minutes:** Duration guard rejects >3600s before download. Chunking never sees >60 min. Videos 20–60 min should chunk, but AudioSegment NameError currently blocks that path.
- 2. **PDF relevance check fails gracefully:** YES — returns True on exception.
- 3. **DB cache when metadata fetch fails:** YES — regex fallback extracts video_id.
- 4. **Grammar fix falls back to raw transcript:** YES — except returns original.
- 5. **Word generation handles malformed markdown:** Partially — exceptions return b ' ' (no crash), but UI does not warn that download is empty.

5. Recommended Fix Order

Fix in this order to restore a working Render deployment and prevent silent data loss:

#	Action	Why first
1	Rewrite _split_audio() with ffmpeg subprocess; remove AudioSegment	Unblocks all caption-less videos; required before any other audio work
2	Wire render.yaml buildCommand to .render-build.sh	Ensures ffmpeg exists on Render for yt-dlp + splitting
3	Relax post-download recovery size check (<24MB gate)	Lets large recovered files reach chunking instead of retrying forever
4	Cap or chunk _fix_grammar for long transcripts	Prevents silent truncation of 40–60 min transcripts
5	Update Railway → Render copy in app.py warnings	Stops user confusion on env var setup
6	Bypass cache when PDF is uploaded; warn on empty docx	Corrects PDF/cache and download UX
7	Align captions URL regex; optional SSL for Postgres; clean Railway leftovers	Reliability polish and repo clarity

End of report. Do not apply fixes from this document until explicitly requested — this review is diagnostic only.